

2(a)	0	B1
2(b)	$pV = nRT$ $(n =) 1.5 \times 10^5 \times 4.2 \times 10^{-3} / 8.31 \times 540$	C1
	= 0.14 mol	A1
2(c)	missing pressure 1.5 ($\times 10^5$)	B1
	both missing volumes 1.8 ($\times 10^{-3}$)	B1
2(d)(i)	(ΔU :) increase in internal energy (of the system)	B1
	(q :) thermal energy supplied to the system	B1
	(W :) work done on system	B1

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Question	Answer	Marks
2(d)(ii)	volume increases and work is done by the gas	B1
	temperature decreases and internal energy decreases	B1

Question	Answer	Marks
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